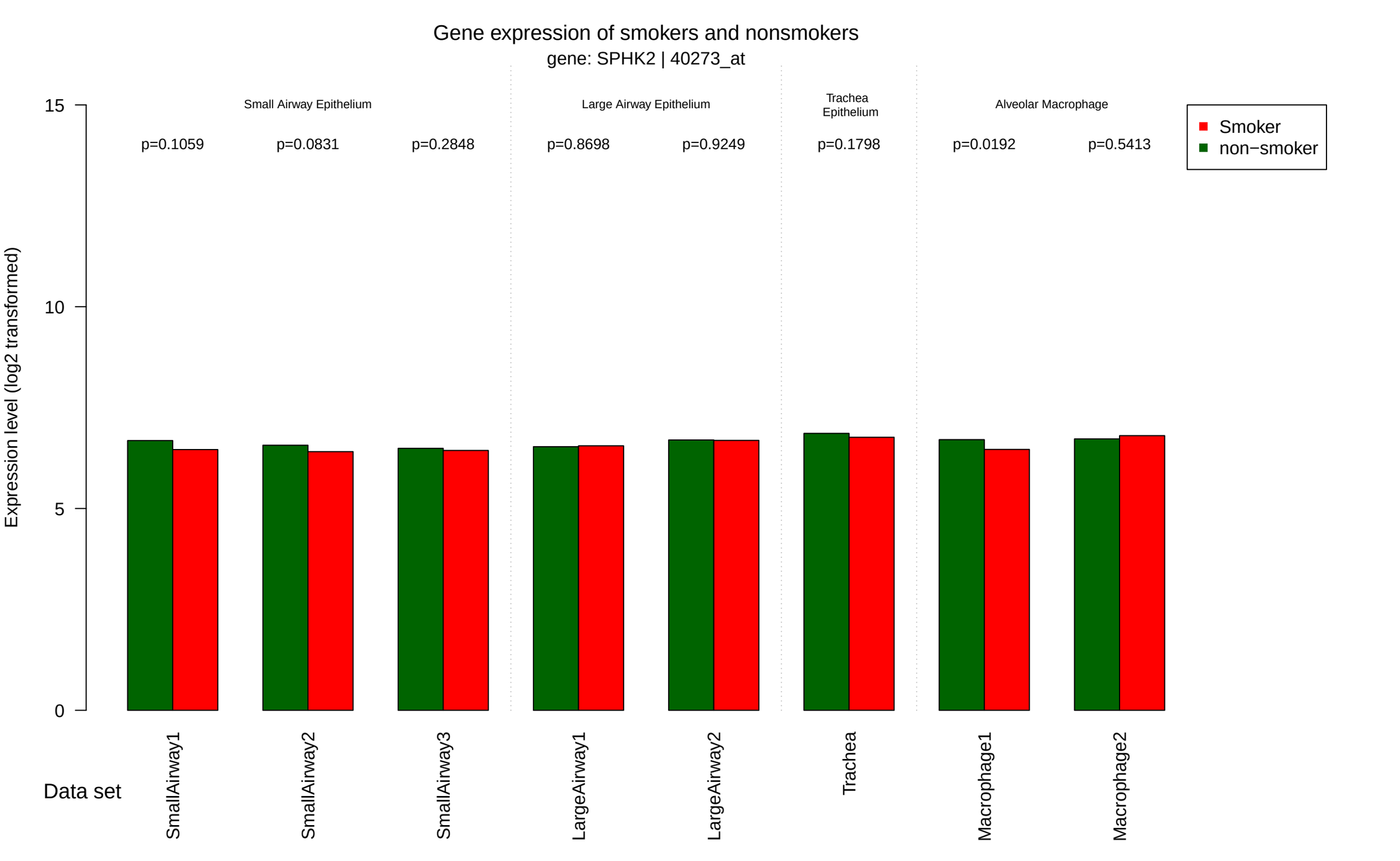
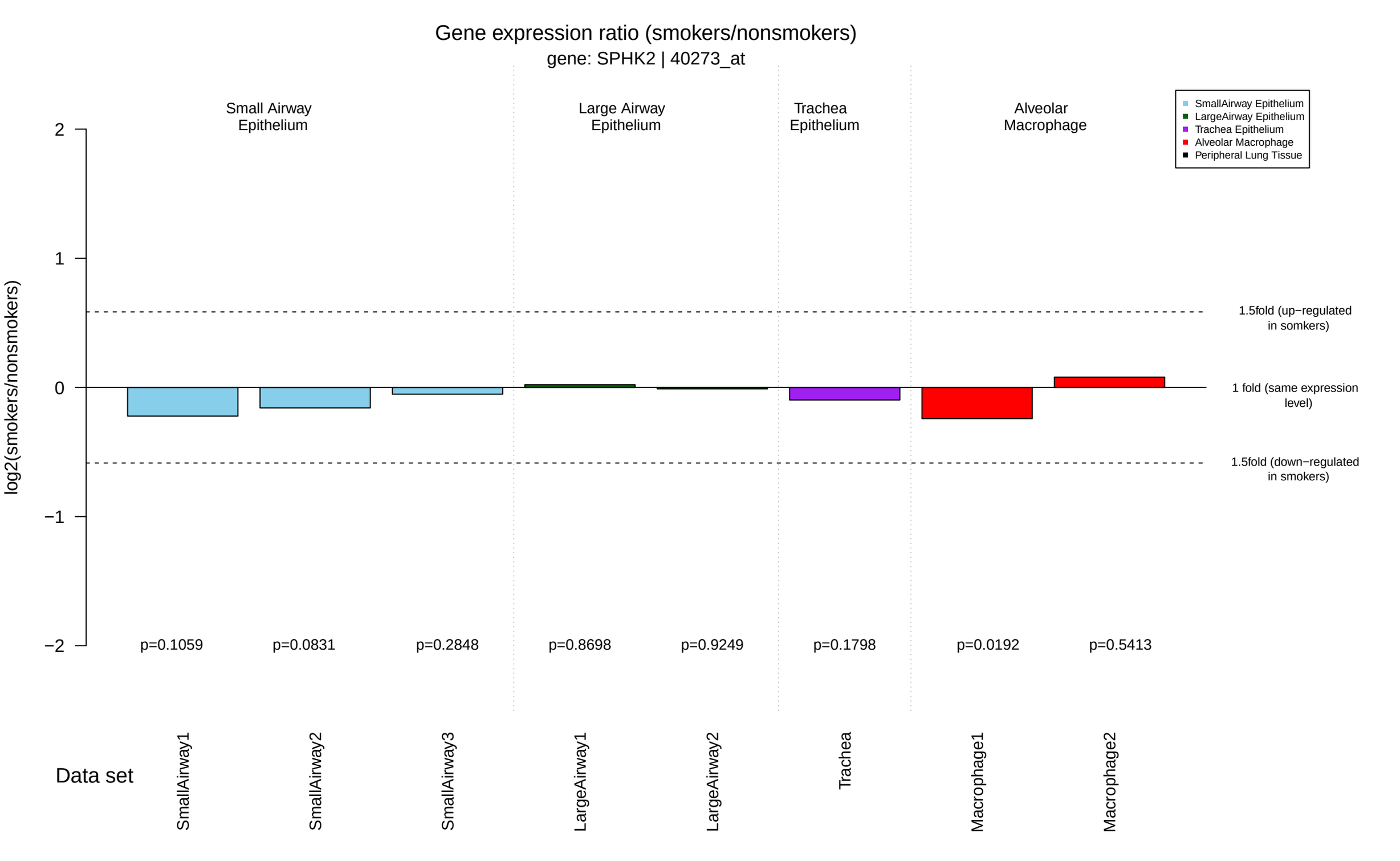
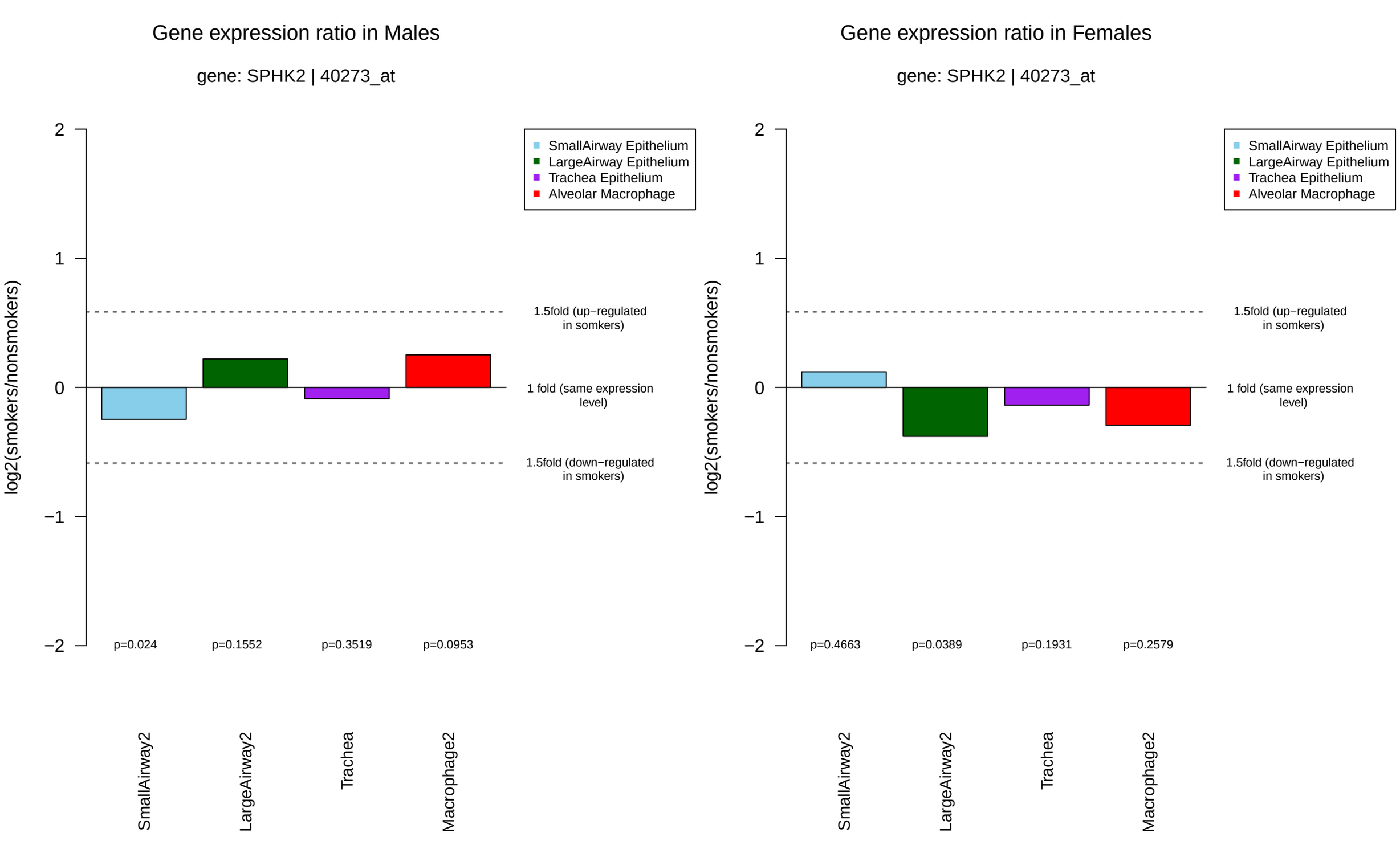
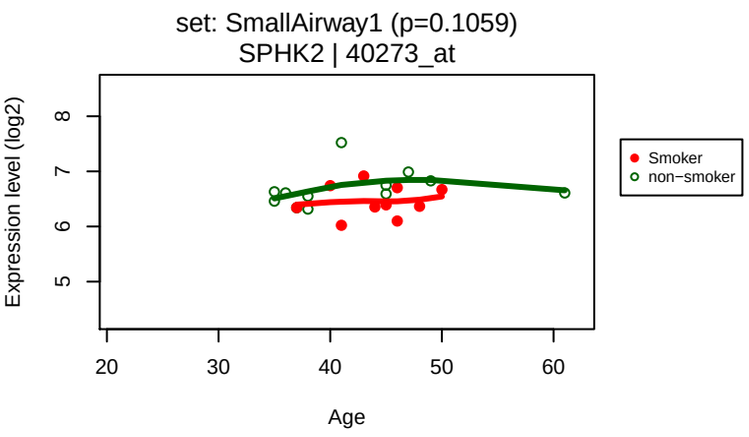
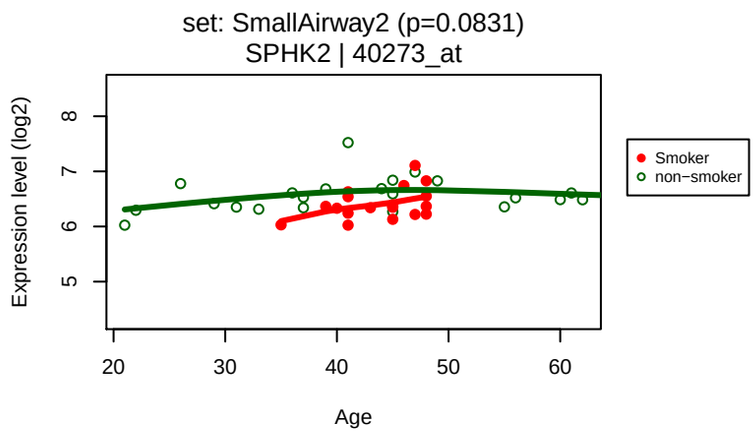
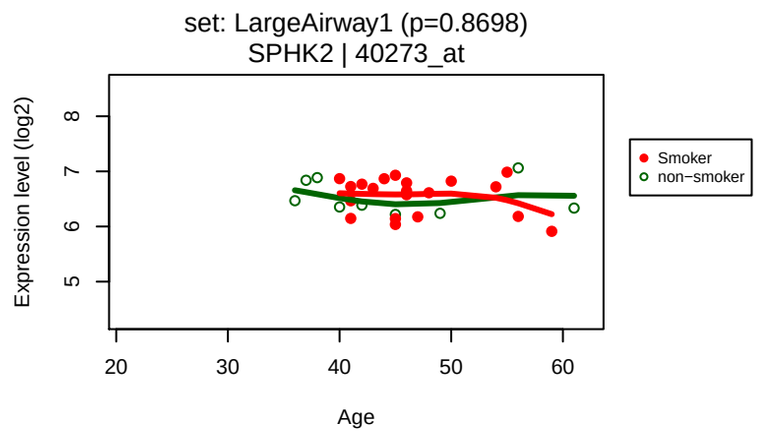

Gene expression of smokers and nonsmokers
gene: SPHK2 | 40273_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1059
p=0.0831
p=0.2848
p=0.8698
p=0.9249
p=0.1798
p=0.0192
p=0.5413
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

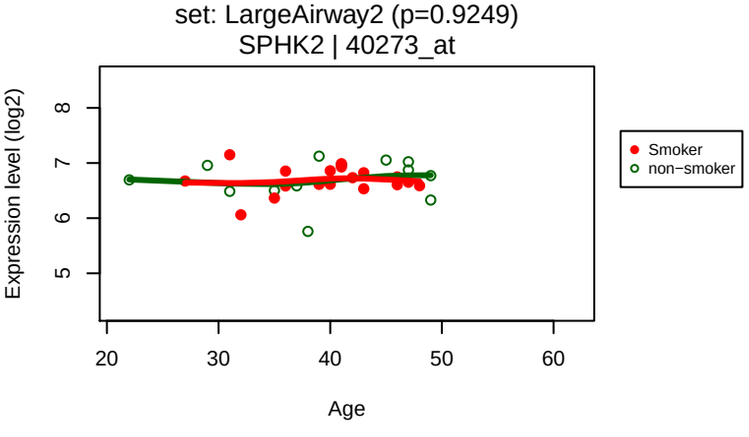

Gene expression ratio (smokers/nonsmokers)
gene: SPHK2 | 40273_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1059
p=0.0831
p=0.2848
p=0.8698
p=0.9249
p=0.1798
p=0.0192
p=0.5413
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

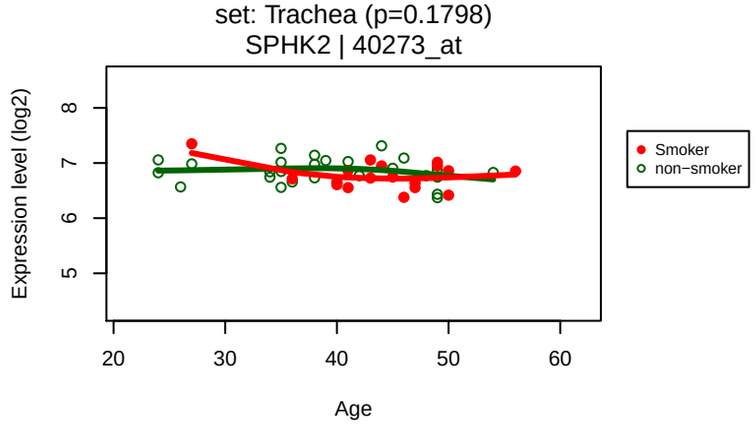

Gene expression ratio in Males
gene: SPHK2 | 40273_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.024
p=0.1552
p=0.3519
p=0.0953
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPHK2 | 40273_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4663
p=0.0389
p=0.1931
p=0.2579
SmallAirway2
LargeAirway2
Trachea
Macrophage2

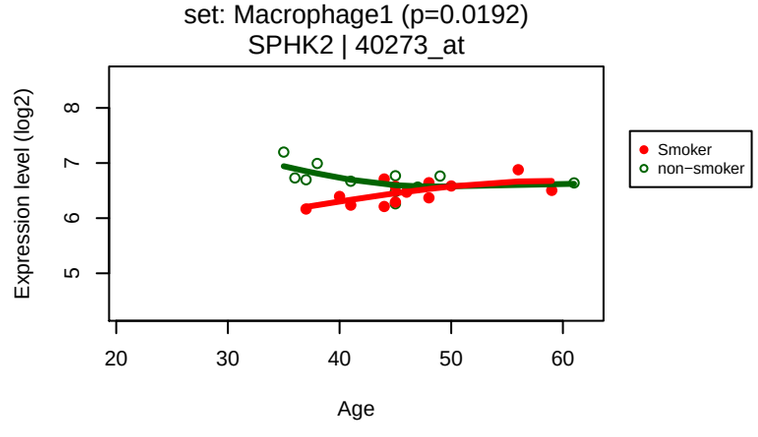

set: SmallAirway1 (p=0.1059)
SPHK2 | 40273_at
Expression level (log2)
Age
Smoker
non−smoker

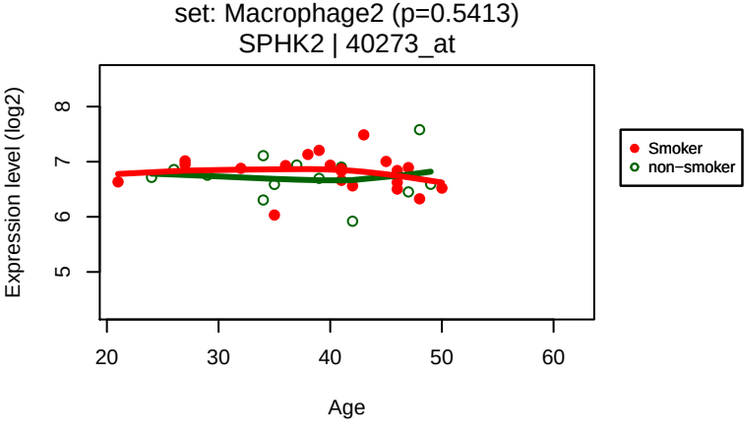

set: SmallAirway2 (p=0.0831)
SPHK2 | 40273_at
Expression level (log2)
Age
Smoker
non−smoker

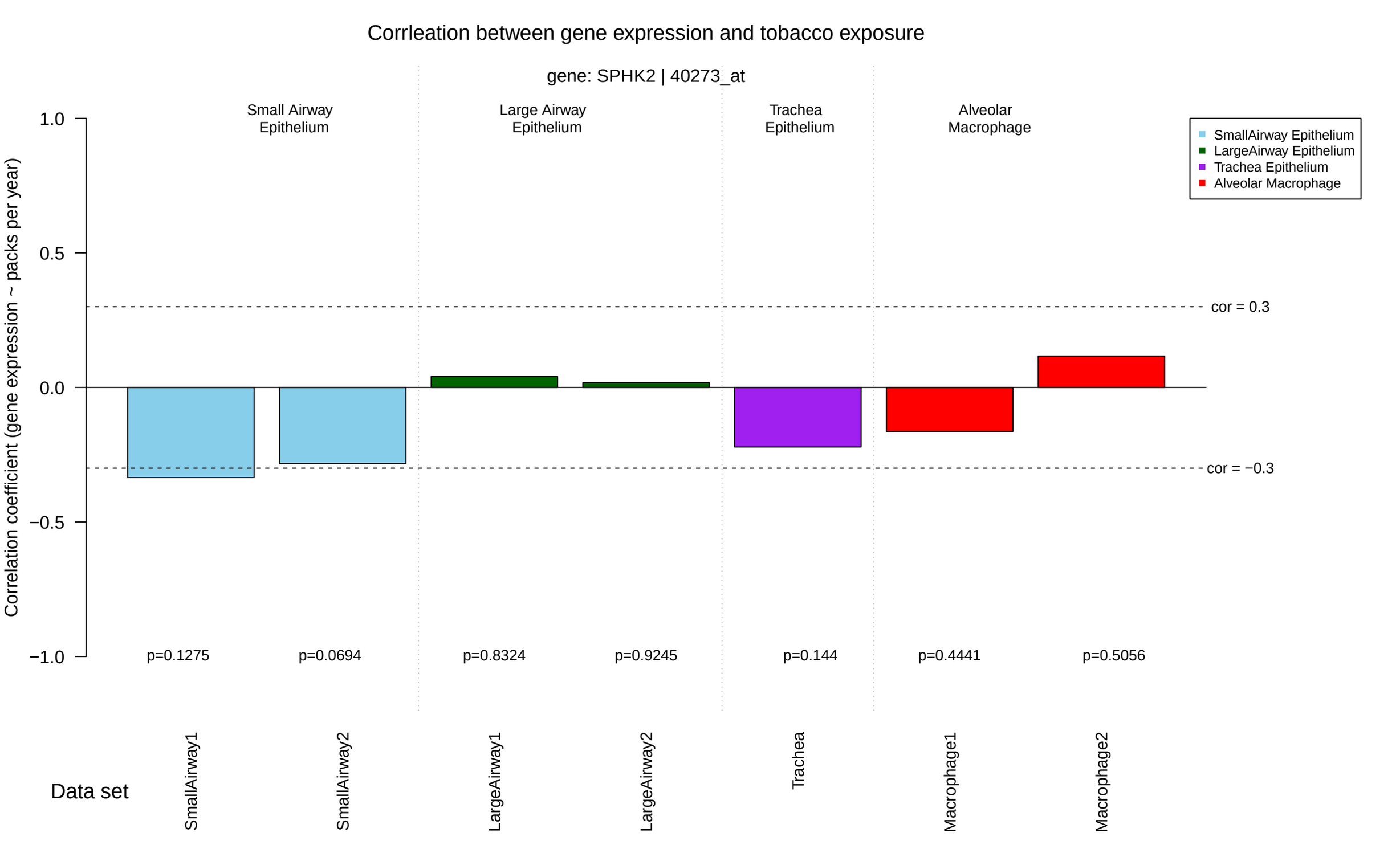

set: LargeAirway1 (p=0.8698)
SPHK2 | 40273_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9249)
SPHK2 | 40273_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1798)
SPHK2 | 40273_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0192)
SPHK2 | 40273_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5413)
SPHK2 | 40273_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPHK2 | 40273_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1275
p=0.0694
p=0.8324
p=0.9245
p=0.144
p=0.4441
p=0.5056
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2